EXAMINATION NO.: _____

MULANJE CLUSTER 2022 MSCE MOCK EXAMINATIONS

CHEMISTRY

Subject Number: M31/I

Tuesday, 31 May

Time allowed: 2 hour
8:00 - 10am

PAPER I

(100 Marks)

Instructions

1. **This paper contains 11 printed pages. Please check.**
2. Fill in your **examination number** at the top of each page.
3. This paper contains two sections, **A** and **B**. In section **A** there are ten short answer questions while in section **B** there are three restricted essay questions.
4. Answer **all** the **thirteen** questions in the spaces provided.
5. Use of electronic calculators is allowed.
6. The maximum number of marks for each answer is indicated against each question.
7. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
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8			
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11			
12			
13			

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Section A (70 marks)

Answer **all** the **ten** questions in this section

1. a. What is a molecule?

_____

_____ **(1 mark)**

b. State **one** factor that affect solubility of a solute in a solvent.

_____

_____ **(1 mark)**

c. Write the formula of a compound formed between potassium ions, K^+ and sulphate ions, SO_4^{2-}.

_____ **(1 mark)**

d. Given $^{7}_{4}Y$ and $^{8}_{4}Y$ as two isotopes of an element Y.

i. Suggest **one** chemical property of element **Y**

_____

(1 mark)

ii. Write an ion formed when element **Y** loses its outermost electron(s).

_____

(1 mark)

iii. Give a reason why the two isotopes react in the same way.

_____

_____

(1 mark)

2. a. Write a balanced chemical equation that describes reversible reaction of hydrated copper sulphate ($CuSO_4*5H_2O$).

(3 marks)

b. A hydrated Magnesium sulphate ($MgSO_4$*$\mathbf{n}H_2O$) was heated to become anhydrous, its mass dropped from 246.5g to 120.5g. Work out the value of **n**. (RFM: $MgSO_4$ = 120; and H_2O = 18).

(3 marks)

c. **Figure 1** is a structure of an element that is composed of four atoms to make a molecule.

i. Identify the element with the structure above.

_____

(1 mark)

ii. State **two** allotropes of the element.

_____

_____ **(2 marks)**

iii. Mention **one** agricultural importance of the element.

_____

_____ **(1 mark)**

3. a. What is a reducing agent?

_____

_____ **(1 mark)**

b. Silver nitrate solution react with zinc metal as shown in the chemical equation.

$$\mathbf{AgNO_3}\ (aq) + \mathbf{Zn}\ (s) \longrightarrow \mathbf{Zn(NO_3)_2}\ (aq) + \mathbf{Ag}(s)$$

i. Identify the spectator ion in the reaction.

_____ **(1 mark)**

ii. Write a balanced oxidation half equation for the reaction.

_____ **(2 marks)**

iii. Describe how Zinc nitrate salt produced from this reaction can be obtained and purified.

_____

_____

_____ **(2 marks)**

c. Give **two** apparatus suitable for conducting distillation method of separating mixtures.

_____

_____ **(2 marks)**

4. a. Define molar gas volume.

_____

_____ **(1 mark)**

b. Calculate the volume occupied by 47.6g of ammonia gas (NH_3) gas at r.t.p.? (RAM, N = 14; H = 1 and Molar gas volume r.t.p is 22.4dm^3).

(4 marks)

c. Give **two** types of intermolecular forces.

_____

_____ **(2 marks)**

5. a. State the difference between saturated hydrocarbons and unsaturated hydrocarbons.

_____

_____ **(2 marks)**

b. Given below are below molecular formula of some organic compounds represented by letters **P**, **Q**, **R** and **S**

P. C_3H_6
Q. C_4H_{10}
R. C_4H_9OH
S. CH_3COOH

i. Identify the function group of compound **S**.

_____ **(1 mark)**

ii. Give the name of compound **R**.

_____ **(1 mark)**

iii. Draw the structural formula of compound **P**.

(1 mark)

iv. Write a chemical equation in which compound **P** undergo addition reaction with chlorine molecule.

(2 marks)

v. Draw and name any **two** isomers of compound **Q**.

(4 marks)

6. a. Give **two** properties which can be used to identify purity of a substance.

_____

_____ **(2 marks)**

b. **Figure 2** below shows a flow diagram of soap preparation.

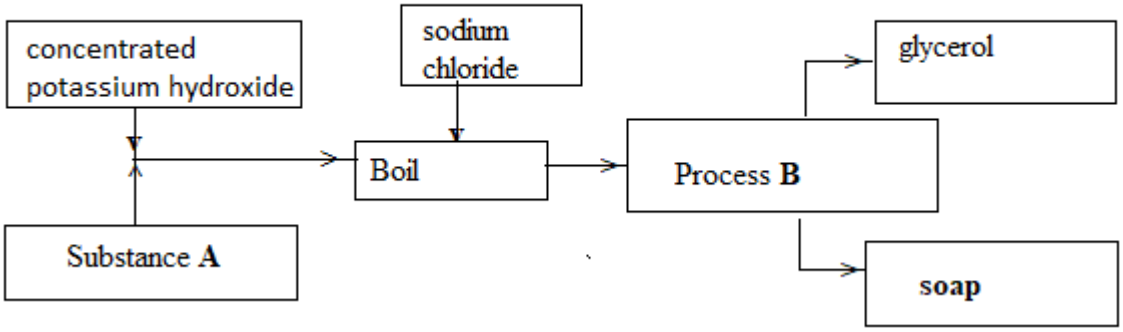

Figure 2

i. Name the homologous series of substance A.

_____ **(1 mark)**

ii. Why is process **B** necessary in soap making process?

_____

_____ **(1 mark)**

iii. Give **one** reason why sodium chloride is added in the process?

_____

_____ **(1 mark)**

7. a. **Figure** 3 is a diagram for Plastics.

X **Y**

Figure 3

i. Name the type of plastic plastics labelled.

X._____ **(1 mark)**

Y._____ **(1 mark)**

ii. State **one** difference between plastic X and plastic Y.

_____

_____

_____

(2 marks)

b. Mention **two** disadvantages of landfill as a waste treatment and disposal method.

_____

_____

_____

(2 marks)

c. Discuss **one** unwanted effects of alkanes.

_____

_____

(2 marks)

d. Explain the importance of the ozone layer.

_____

_____

_____

(2 marks)

8. a. Give **one** chemical property of acidic oxide.

_____

_____

(1 mark)

b. Describe how the presence of a catalyst does increases rate of reaction?

_____

_____

(2 marks)

9. a. What does ΔH_n^o denotes?

_____

_____

(1 mark)

b. Nitrogen gas react with hydrogen gas to produce ammonia gas as shown in the equation:

$$N_2 + 3H_2 \rightarrow 2NH_3$$

Given the bond energy values as shown in table **2** below

Bond	**Energy ($kjmol^{-1}$)**
N-H	390
N≡N	942
H-H	435

Table 2

i. Calculate the overall energy changes for the reaction above.

(4 marks)

ii. Determine the type of heats of reaction for the reaction above.

_____

(1 mark)

10. a. Identify a general trends of the periodic table that makes atoms to attract electrons from other atoms.

_____

(1 mark)

b. Draw a dot and cross diagram for hydrogen chloride (HCl).

(3 marks)

Section B (30 marks)

Answer **all** the **three** questions in this section.

11. a. Describe how carbonate ions are used to remove lead ions (Pb^{2+}) in industrial effluents before releasing them into water bodies.

_____

_____

_____

_____

_____

_____

_____

(**4 marks)**

b. Discuss **three** benefits of recycling metals and plastics.

_____
_____
_____
_____
_____
_____
_____

(6 marks)

12. a. Describe **two** industrial importance of water.

_____
_____
_____
_____

(4 marks)

b. With the aid of chemical equations, describe the process of rusting.

_____
_____
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(6 marks)

13. With the aid of a well labelled diagram, describe the water cycle.

(10 marks)

END OF QUESTION PAPER

NB: This paper contains 11 printed pages.